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328452(28)

B. E. (Fourth Semester) Examination,
Nov.-Dec. 2015

(New Scheme)

(Et. & T Engg.)

ANALOG COMMUNICATION

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all questions. Part (a) of each question is compulsory and carries 2 marks each. Attempt any two parts from (b), (c) and (d) and carries 7 marks each.

(a) Show that the squaring circuit will not permit the generation of a local oscillator signal capable of demodulating an SSB-SC sigand.

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(b) Explain the working of a diode-demodulator/envelope detector. Specify the conditions in terms of modulation index so that the message signal can be perfectly recovered from the AM signal using this method.

(c) The signal

$$v(t) = \left[1 + 0 \cdot 2 \cos\left(\omega_M / 3\right)t\right] \cos \omega_c t$$

is demodulated using a square-law demodulator having the characteristic $v_0 = v^2$. The output $v_0(t)$ is then filtered by an ideal low-pass filter having a cut-off frequency at f_M Hz. Sketch the amplitude frequency characteristics of the output waveform in the frequency range $0 <= f <= f_M$.

(d) A baseband signal, bandlimited to the frequency range 300-3000 Hz, is to be superimposed on a carrier of frequency 40 MHz as a single-sideband modulation using the filter method. Assume that filters are available which will provide 40 dB of attenuation in a frequency interval which is about 1 percentage of the filter center frequency. Draw a block diagram of the suitable system.

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(a) Distinguish between Narrow band FM and wide band FM.

(b) Explain the direct method of frequency modulation in details.

(c) Show that $v(t)$ given by :

$$v(t) = \cos\left(2\pi 10^6 t\right) + 0{\cdot}02\cos\left[2\pi\left(10^6 + 10^3\right)t\right]$$

represents a carrier which is modulated both in amplitude and frequency by drawing a phasor diagram. Find value of m and β.

(d) In an Armstrong modulator the crystal oscillator frequency is 200 kHz. It is desired in order to avoid distortion, to limit the maximum angular deviation to $\phi_m = 0{\cdot}2$. The system is to accommodate modulation frequencies down to 40 Hz. At the output of modulator the carrier frequency is to be 108 MHz & the frequency deviation 80 kHz. Select multiplier & mixer oscillator frequencies to accomplish this end.

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3. (a) Define White Noise.

(b) Explain the effect of filtering on the probability density of Gaussian Noise.

(c) The auto-correlation function of noise signal is triangular and defined as :

$$R_n(\tau) = \begin{cases} 1-|\tau| & \text{for } |\tau| < 1 \\ 0 & \text{for } |\tau| > 1 \end{cases}$$

Find its noise spectrum.

(d) Find the effect of Linear filtering of White Noise by filtering through a RC low pass filter

4. (a) Define the figure of Merit.

(b) Determine the signal to noise ratio for an AM system using a Square Law Demodulator.

(c) An audio signal of 4 kHz Bandwidth is to be transmitted through a channel that introduces 30 dB loss and white noise of PSD 10 [illegible] W/Hz. Calculate the minimum required transmitted power if the message is sent by SSB-SC, DSB-SC and DSB-

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modulation methods. The received output SNR should be at least 40 dB. For DSB-C energy in the sideband is half of that of the carrier.

(d) The time average of the square of a modulating message signal of 60 kHz bandwidth is calculated as 0.1 W. The signal is used in DSB-C modulation with carrier power 10 W. If the additive white noise PSD is 10^{-9} W/Hz. Find output SNR for a square law demodulator. Also find the output SNR if the carrier power is reduced by 100 times.

5. (a) What is an Amplitude Limiter? Explain its significance.

(b) Explain the need and effect of pre-emphasis and de-emphasis networks used in commercial FM systems.

(c) Find the output SNR of an FM limiter-demodulator when input signal strength is 0.5 W, maximum frequency deviation 60 kHz, baseband signal cut-off frequency 15 kHz, received white Gaussian noise PSD 10^{-10} W/Hz. And average power of the modulating signal 0.1 W. Find the required

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transmitted power for above if channel has 26 dB loss and required output SNR is at least 40 dB.

(d) Compare FM and AM in terms of the figure of merit.

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